

**Glowing frogs!**

Many species of frog and salamander glow under certain wavelengths of light. <https://digit.in/mar20-53>

**Sandy side of the Moon**

Chinese rover Yutu-2 discovered 130ft of regolith, a fine, dusty environment, on the far side of the Moon. <https://digit.in/mar20-54>



Exploring Saturn's Moons

The mysteries of the Saturnian moons have only deepened after the Cassini mission

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aturn is romanticized as the “Jewel of the Solar System” and is a spectacular planet indeed with its rings.

While the Pioneer and Voyager spacecraft executed flybys of the Saturnian system in the late 1970s and early 1980s, these low resolution images revealed little information about the gas giant and the ice moons that orbit it. The flagship Cassini-Huygens mission, a collaboration between NASA and ESA was the most fruitful and informative explorer of Saturn and its moons. The data gathered by Cassini revealed some surprising things that scientists had no idea about. Enceladus, had water spouts that sprang out of its subsurface ocean. The discovery made scientists change the trajectory of Cassini to better understand these spouts. The finding, along with other measurements indicates that Enceladus has all the ingredients

necessary to support life. Titan, another interesting moon has hydrocarbon lakes on the surface, that could provide the basis of an exotic form of biochemistry. It is the only other body in the Solar System with a liquid level as well as a dense atmosphere. Hiding beneath the surface, maybe a subsurface liquid water ocean as well, containing life as we know it on Earth. Cassini discovered that the atmosphere of Titan was teeming with molecules that indicated complex chemistry. The surface may in fact be a laboratory of prebiotic conditions, and may one day help us better understand how life originated on Earth. Cassini also gathered data to indicate that both Dione and Rhea may have subsurface oceans as well. However, the most promising targets for future exploration, especially when it comes to finding life elsewhere in the Solar System, are Enceladus and Titan. We cannot know more about these remote ice moons without sending follow up probes, with the specific intent of discovering any life forms. Scientists simply had

no clue about these potential niches for life when Cassini was originally designed. While there have been concepts for future missions to the Saturnian moons, there are none that are scheduled to reach their destinations for the next decade and a half. For a number of reasons, missions to other bodies have been picked over the proposals for follow up missions to Saturn. These include the OSIRIS-REx mission to Asteroid Bennu, the Juno mission to Jupiter, and the New Horizons mission to the outer Solar System. The data gathered by the Cassini mission continues to be analysed, and serves as the basis for research that is expected to continue for decades.

EXPLORING THE HYDROCARBON SEAS

The Titan Mare Explorer TIME is a concept proposal that is meant to specifically investigate the discovery of liquid bodies on the surface of Titan. The bodies of hydrocarbon liquids are large enough to support a cycle, similar to the water cycle on Earth. It